

Influence diagrams as attributed C-sets

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1 Overview

An influence diagram ([Howard and Matheson 2005](#)) extends a Bayesian network with decisions, the information available when each decision is taken, utility functions and a decision order (SPEC section 23). `InfluenceDiagrams.jl` represents it as an attributed C-set over the schema `SchInfluenceDiagram`, which extends the `SchBayesNet` schema of `BayesianNetworks.jl` with five objects:

object	meaning	homs
Decision	a decision controlling an action variable	decision_variable
InformationInput	one variable known when a decision is taken	information_decision, information_variable
Utility	a utility node	
UtilityInput	one variable read by a utility node	utility_node, utility_variable
DecisionPrecedence	an explicit ordering of two decisions	earlier, later

The three kinds of arc of SPEC section 23 (causal, informational, value) are three different objects, so they can never be confused: a causal Input feeds a mechanism, an InformationInput feeds a decision, a UtilityInput feeds a utility node. The representation choices this settles – separate value nodes, information arcs that are not causal, an explicit decision order – are the ones surveyed by Bielza et al. (2011).

Two conditions on the decision order are worth keeping apart from the start. Shachter (1986) calls a diagram *regular* when a directed path runs through all the decisions; *no-forgetting* (perfect recall) is the stronger property that everything known at a decision is still known at every later one. `validate` requires neither, because a diagram that forgets is a well-formed limited-memory influence diagram (Lauritzen and Nilsson 2001).

2 Setup

```
using InfluenceDiagrams
using Graphs # 'ne' counts the edges of the 'SimpleDiGraph's returned below
```

3 The schema

```
SchInfluenceDiagram
```

```

ACSets.Schemas.BasicSchema{Symbol}([:Variable, :State, :Mechanism, :Input,
:Decision, :InformationInput, :Utility, :UtilityInput, :DecisionPrecedence],
[(:state_variable, :State, :Variable), (:target, :Mechanism, :Variable),
(:input_mechanism, :Input, :Mechanism), (:input_variable, :Input, :Variable),
(:decision_variable, :Decision, :Variable), (:information_decision,
:InformationInput, :Decision), (:information_variable, :InformationInput,
:Variable), (:utility_node, :UtilityInput, :Utility), (:utility_variable,
:UtilityInput, :Variable), (:earlier, :DecisionPrecedence, :Decision), (:later,
:DecisionPrecedence, :Decision)], [:Label, :Position, :Ref], [(:variable_name,
:Variable, :Label), (:space_ref, :Variable, :Ref), (:state_name, :State, :Label),
(:state_position, :State, :Position), (:mechanism_name, :Mechanism, :Label),
(:kernel_ref, :Mechanism, :Ref), (:input_position, :Input, :Position),
(:decision_name, :Decision, :Label), (:utility_name, :Utility, :Label),
(:utility_ref, :Utility, :Ref), (:information_position, :InformationInput,
:Position), (:utility_position, :UtilityInput, :Position)], Tuple{Union{Nothing,
Symbol}, Symbol, Symbol, Tuple{Tuple{Vararg{Symbol}}, Tuple{Vararg{Symbol}}}}{[]})

```

The same schema is defined in the Lean project under `proofs/`, which emits `proofs/schemas/influence_diagram`. The test suite checks that `schema_json(SchInfluenceDiagram)` agrees with it.

```

sj = schema_json(SchInfluenceDiagram)
[d["name"] for d in sj["Ob"]]

```

```

9-element Vector{String}:
 "Variable"
 "State"
 "Mechanism"
 "Input"
 "Decision"
 "InformationInput"
 "Utility"
 "UtilityInput"
 "DecisionPrecedence"

```

4 Building a diagram

`influence_diagram` mirrors `bayesnet`: variables with their states, mechanisms with ordered parents, decisions with ordered information sets and utilities with ordered scopes. The umbrella problem has a chance variable `Weather`, a forecast that depends on it, a decision `Umbrella` that sees the forecast, and a utility over the weather and the action.

```

id = influence_diagram(:Weather ⇒ [:sunny, :rainy],
                      :Forecast ⇒ [:sunny, :cloudy, :rainy],
                      :Umbrella ⇒ [:take, :leave];
                      mechanisms = [:Forecast ⇒ :Weather],
                      decisions = [:Umbrella ⇒ :Forecast],
                      utilities = [:U ⇒ (:Weather, :Umbrella)])

```

Variable	variable_name	space_ref
1	Weather	NoRef()
2	Forecast	NoRef()
3	Umbrella	NoRef()

State	state_variable	state_name	state_position
1	1	sunny	1
2	1	rainy	2
3	2	sunny	1
4	2	cloudy	2
5	2	rainy	3
6	3	take	1
7	3	leave	2

Mechanism	target	mechanism_name	kernel_ref
1	2	Forecast_mechanism	NoRef()
2	1	Weather_mechanism	NoRef()

Input	input_mechanism	input_variable	input_position
1	1	1	1

Decision	decision_variable	decision_name
1	3	Umbrella

InformationInput	information_decision	information_variable	information_position
1	1	2	1

Utility	utility_name	utility_ref
1	U	NoRef()

UtilityInput	utility_node	utility_variable	utility_position
1	1	1	1
2	1	3	2

Every object is inspectable. Decisions are named after their action variable unless told otherwise, and the information set is returned in `information_position` order, the domain order of the decision's policy.

```
decision_names(id), information_names(id, :Umbrella), utility_scope_names(id, :U)

([:Umbrella], [:Forecast], [:Weather, :Umbrella])
```

```
action_names(id), chance_names(id)

([:Umbrella], [:Weather, :Forecast])
```

5 Information arcs are not causal arcs

The forecast informs the umbrella decision, but nothing generates the umbrella: the action variable has no mechanism and no parents. The derived causal graph (`variable_graph`) only contains `Weather → Forecast`.

```
has_mechanism(id, :Umbrella), parents(id, :Umbrella)

(false, Int64[])
```

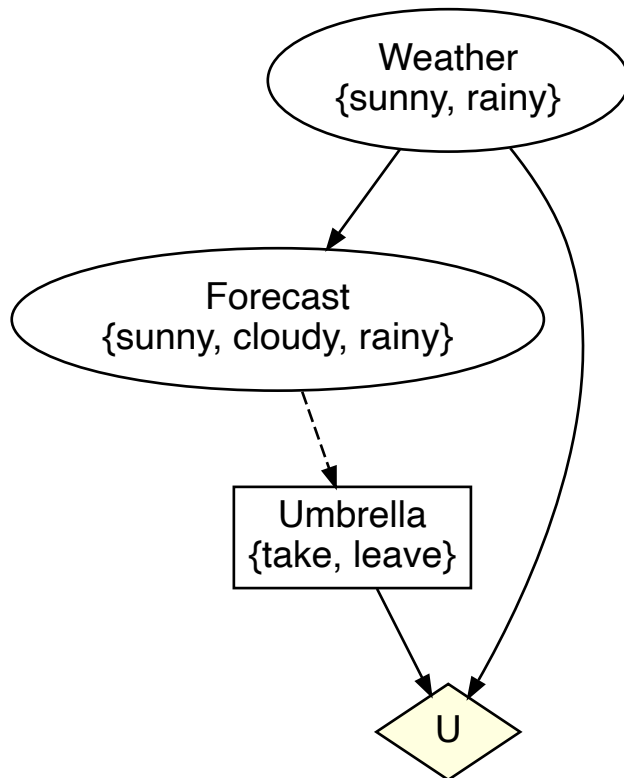
`information_graph` adds the dashed arcs (information variable to action) and the explicit precedence; the declared information is consistent when it is acyclic, and `decision_order` reads the order of the decisions off it. Acyclicity is weaker than *no-forgetting* (perfect recall), which decision variable elimination needs and which `no_forgetting_arcs` and `with_no_forgetting_report` and `repair`; vignette 04 covers it.

```
ne(variable_graph(id)), ne(information_graph(id))

(1, 2)
```

`to_graphviz` draws all three kinds of arc: chance variables are ellipses, decisions are boxes, utilities are diamonds, and information arcs are dashed.

```
to_graphviz(id)
```



6 Utilities are not variables

A utility node has no states and cannot be the parent of anything, because it is not a `Variable`; SPEC section 37 item 5 (“utility nodes have no children”) holds by construction.

```
has_variable(id, :U), nparts(id, :Utility), nparts(id, :UtilityInput)
```

```
(false, 1, 2)
```

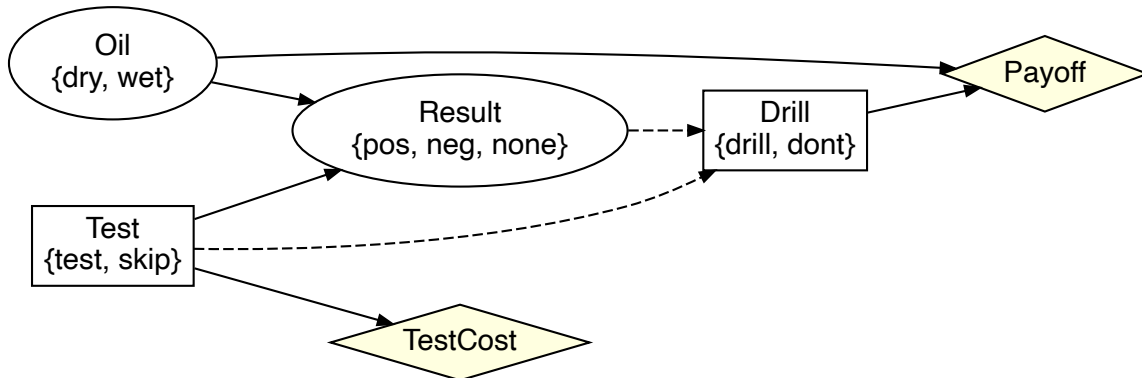
7 Precedence and sequential decisions

With several decisions, the order is induced by the arcs: an action that informs (or causally affects an information variable of) another decision precedes it. The two-stage test-then-act example of `two_stage_diagram` has `Test` before `Drill` because `Drill` observes `Test` and the test result.

```
t = two_stage_diagram()
decision_name.(Ref(t), decision_order(t))
```

```
2-element Vector{Symbol}:
 :Test
 :Drill
```

```
to_graphviz(t; rankdir = "LR")
```



Unordered decisions can be ordered explicitly with `add_precedence!` (or the `precedence` keyword); a precedence that contradicts the arcs is a validation error.

```
add_precedence!(t, :Drill ⇒ :Test)
try
    validate(t)
catch e
    e
end
```

```
DecisionPrecedenceCycleError([:Test, :Drill])
```

8 Validation

`validate` runs the structural checks of `BayesianNetworks.jl` on the chance part and the twelve checks of SPEC section 37 on the decision part, throwing a typed error for the first violation; `validation_errors` lists all of them. With `closed = true` every variable must have exactly one generator: a chance mechanism or a decision.

```
validate(id; closed = true)
```

An information variable downstream of the action is not available when the decision is taken:

```
bad = influence_diagram(:Weather ⇒ [:sunny, :rainy], :Umbrella ⇒ [:take, :leave],
    :Wet ⇒ [:no, :yes];
    mechanisms = [:Wet ⇒ (:Weather, :Umbrella)],
    decisions = [:Umbrella ⇒ :Wet], utilities = [:U ⇒ :Wet])
validation_errors(bad)
```

```
1-element Vector{Exception}:
 InvalidInformationSetError(:Umbrella, :Wet, :downstream)
```

An action variable with a chance mechanism, or controlled by two decisions, violates decision uniqueness:

```
dup = umbrella_diagram()
add_mechanism!(dup, :Umbrella; inputs = [:Forecast])
validation_errors(dup)
```

```
1-element Vector{Exception}:
 DecisionUniquenessError(:Umbrella, :Umbrella, :has_mechanism)
```

9 From and to Bayesian networks

An `InfluenceDiagram` is also an `AbstractBayesNet`, so every inspection function of `Bayesian-Networks.jl` applies to it, and the two types convert into each other by copying the shared objects (the pullback data migration along the schema inclusion). `BayesNet(id)` drops the decision layer and leaves the action variables without a mechanism, an open network; `InfluenceDiagram(bn)` is the starting point for adding decisions to an existing network.

```
bn = BayesNet(id)
nparts(bn, :Mechanism), has_mechanism(bn, :Umbrella)
```

```
(2, false)
```

```
id2 = InfluenceDiagram(bn)
add_decision!(id2, :Umbrella; information = [:Forecast])
add_utility!(id2, :U; scope = [:Weather, :Umbrella])
is_isomorphic(id2, id)
```

```
true
```

Diagrams serialise to JSON in the same envelope as networks (`json_influence_diagram`), and `canonicalize` rennumbers parts deterministically so that diagrams built in different orders compare equal.

10 Summary

An influence diagram is one `ACSet` over `SchInfluenceDiagram`, so decisions, information sets, utilities and the decision order are data of the same structure as the underlying Bayesian network, and the three kinds of arc are three different schema objects rather than a convention. `validate` checks what makes the diagram well-formed – acyclicity, resolvable references, consistent information sets – and deliberately not no-forgetting, and `BayesNet/InfluenceDiagram` move

between the two schemas without loss. The next vignette, *Policies and expected utility*, binds numbers to this syntax and computes what a strategy is worth.

11 References

- Bielza, Concha, Manuel Gómez, and Prakash P. Shenoy. 2011. "A Review of Representation Issues and Modeling Challenges with Influence Diagrams." *Omega* 39 (3): 227–41. <https://doi.org/10.1016/j.omega.2010.07.003>.
- Howard, Ronald A., and James E. Matheson. 2005. "Influence Diagrams." *Decision Analysis* 2 (3): 127–43. <https://doi.org/10.1287/deca.1050.0020>.
- Lauritzen, Steffen L., and Dennis Nilsson. 2001. "Representing and Solving Decision Problems with Limited Information." *Management Science* 47 (9): 1235–51. <https://doi.org/10.1287/mnsc.47.9.1235.9779>.
- Shachter, Ross D. 1986. "Evaluating Influence Diagrams." *Operations Research* 34 (6): 871–82. <https://doi.org/10.1287/opre.34.6.871>.